ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) Path to success

ROADMAP PROBLEM : 11 SOLUTION

A $CH_3 - CH_2 - COOH$

B $CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - COOH$ HVZ reaction

C $CH_3 - \underset{\displaystyle NH_2}{\underset{|}{CH}} - COOH$ Alanine

D $CH_3 - CH_2 - NH_2$ Schmidt degradation

E $Ph - SO_2 - NH - Et$

F $CH_3 - CH_2 - OH$ Diazotisation

G $CH_3 - CH_2 - COOEt$ Esterification

H $CH_3 - CH_2 - \overset{\displaystyle O}{\overset{\|}{C}} - \underset{\displaystyle CH_3}{\underset{|}{CH}} - COOEt$ Claisen condensation

I $CH_3 - CH_2 - \overset{\displaystyle O}{\overset{\|}{C}} - CH_2 - CH_3$

J $CH_3 - CH_2 - \overset{\displaystyle O}{\overset{\|}{C}} - Cl$

K $CH_3 - CH_2 - CH_2 - COOH$ Arndt Eistert reaction

L $CH_3 - CH_2 - Br$ Hundsdeicker reaction

M $CH_3 - CH = O$

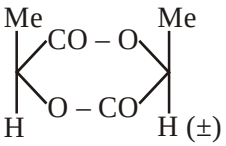
N $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CN$ Cyanohydrin formation

O $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - COOH$ Lactic acid

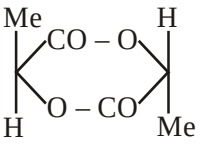
P $CH_3 - \overset{\displaystyle O}{\overset{\|}{C}} - COOH$ Pyruvic acid

Q_1, Q_2 and Q_3 is a mixture of three lactides.

Here it is heating effect of α-hydroxy acid is applied.



Q_1 / Q_2
optically active



Q_3 meso
due to COS

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R CHI_3

S oxalic acid

Formation of R and S is Iodoform reaction

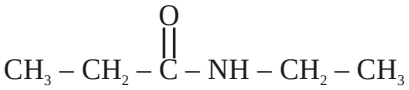
T HCOOH (Heating effect on oxalic acid)

U $HC \equiv CH$

V $H_2C = CH - COOH$ Acrylic acid

W $H_2C = CH - COOMe$ methyl acrylate

This on polymerisation gives polymethyl acrylate (PMA) substance used to make acrylic glasses.

X  Schmidt reaction

Y $CH_3 - CH_2 - NC$ Carbylamine reaction

Z $CH_3 - CH_2 - CH_2 - CH_3$ Kolbe's electrolysis.

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